

Mandibular Major Connectors

Learning objectives:

At the end of this lecture, the student can

- Differentiate between different types of mandibular major connectors.
- Describe the indications and contraindications of each major connector.
- Draw all the types of mandibular major connectors.
- Describe how to measure the height of the floor of the mouth.

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- A. Lingual Bar
 - B. Lingual Plate (Linguoplate)
 - C. Sublingual bar
 - D. Cingulum Bar (Continuous Bar)
 - E. Labial Bar

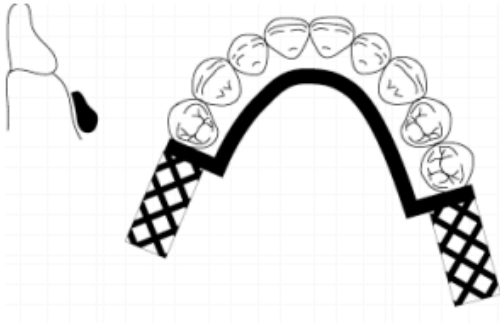
A. Lingual bar

This is the most commonly used mandibular major connector.

Shape and position

- flat on the tissue side
- convex or tear-drop on tongue side (1/2 pear shape, with thin edge toward teeth)
- Occluso-gingival width = 4 to 6 mm.
- Superior border located at least 4 mm inferior to the gingival margins
- The inferior border should be as low as the lingual frenum and tissue reflections of the floor of the mouth will permit, as determined by observing functional movements of the tongue.

Indications: It should be used whenever possible unless advantages can be obtained from another connector. Such situations are rare.



Contraindications:

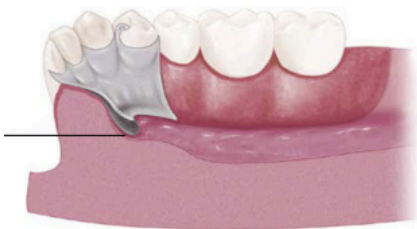
1. When the lingual frenum is high or the space available for a lingual bar is limited (less than 8 mm).
2. Lingually inclined teeth.
3. Lingual undercut in the alveolar ridge, which would result in an excessive space between the bar and the mucosa.
4. When planning for future replacement of one or more incisor teeth.

B. Lingual Plate (Linguoplate)

Shape and position: The lingual plate consists of a lingual bar plus an extension over the gingula of anterior teeth and lingual surfaces of involved posterior teeth at their height of contour.

A lingual plate should be made as thin as is technically feasible and should be contoured to follow the contours of the teeth and the embrasures. The upper border should follow the natural curvature of the supra-cingular surfaces of the teeth and should not be located above the middle third of the lingual surface, except to cover interproximal spaces to the contact points.

A rest area must be prepared at each end of the anterior segments of the lingual plate. Any areas where the connector crosses the gingival tissue must be relieved to avoid impingement caused by the movement of the lingual plate.



Indications

This mandibular major connector should only be considered when a lingual bar cannot be used.

1. high floor of the mouth.
2. prominent lingual frenum.
3. lingual tori that would be impinged upon by a lingual bar
4. Extensive distal extension cases with severe vertical resorption of ridges (i.e. little resistance to horizontal rotational tendencies). In these cases, the major connector could provide some additional stability.
5. When the future replacement of one or more incisor teeth will be facilitated by the addition of retention loops to an existing lingual plate.

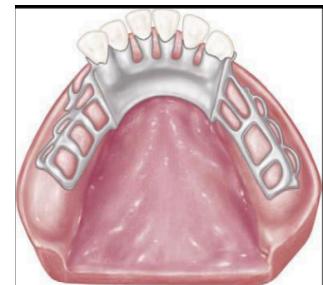


(Patel et al, 2009)¹

Contraindication

1. Crowded anterior teeth, lead to small gaps between the superior edge of the plate and the teeth.
2. Lingually inclined teeth.
3. Open cervical embrasures or large diastema where the plate would be visible.

When anterior teeth are spaced, and the patient objects to metal showing through the spaces. The linguoplate can then be constructed so that the metal will not appreciably show through the spaced anterior teeth (**interrupted lingual plate**) The rigidity of the major connector is not greatly altered. However, such a design may be as much of a food trap.



¹ Patel MB, Bencharit S. A treatment protocol for restoring occlusal vertical dimension using an overlay removable partial denture as an alternative to extensive fixed restorations: a clinical report. The Open Dentistry Journal. 2009 Oct;3:213-218. DOI: 10.2174/1874210600903010213. PMID: 19915723; PMCID: PMC2776309.

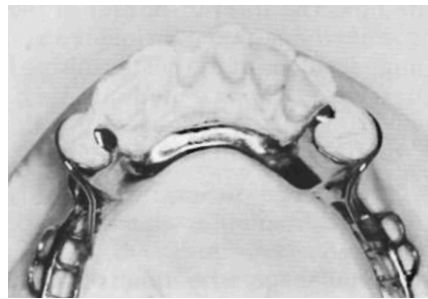
C. Sublingual bar

Shape and position: The bar shape remains essentially the same as that of a lingual bar, but placement is inferior and posterior to the usual placement of a lingual bar, lying over and parallel to the anterior floor of the mouth.

Indications:

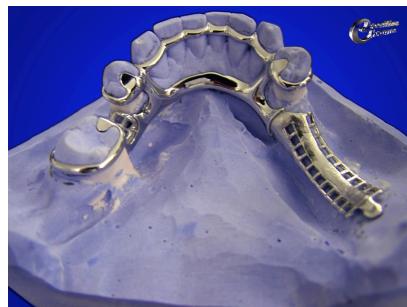
1. When the height of the floor of the mouth does not allow placement of the superior border of the lingual bar at least 4 mm below the free gingival margin.
2. Instead of a lingual plate, if the lingual frenum does not interfere,
3. In the presence of a lingual undercut of the anterior teeth that would require considerable block-out for a conventional lingual bar.

Contraindications: include interfering lingual tori, high attachment of a lingual frenum, and interference with elevation of the floor of the mouth during functional movements.



D. Cingulum Bar (Continuous Bar)

Shape and position: A cingulum bar may be added to the lingual bar or can be used independently. It is a narrow (3 mm) metal strap located on the cingula of anterior teeth, scalloped to follow interproximal embrasures with inferior and superior borders tapered to tooth surfaces. The cingulum bar originates bilaterally from incisal, lingual, or occlusal rests of adjacent principal abutments.

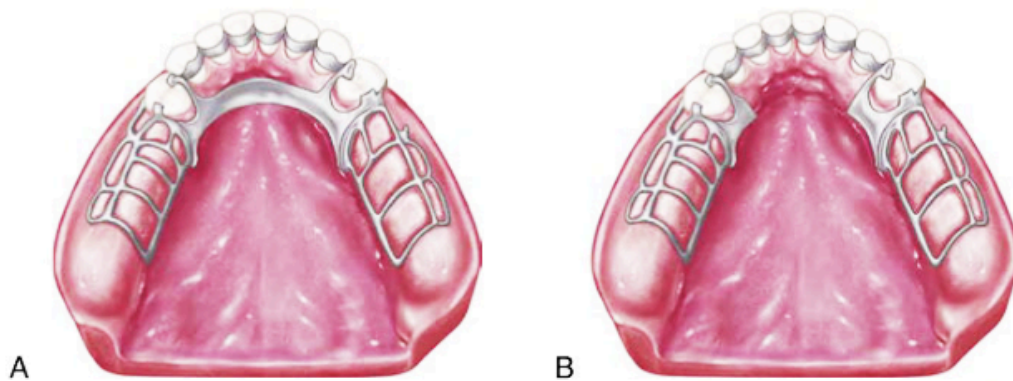


Indications

When a Linguoplate is the major connector of choice but the axial alignment of the anterior teeth is such that excessive block-out of interproximal undercuts must be made, a cingulum bar may be considered.

Contraindications:

1. When a wide diastema exists between the lower anterior teeth.
2. Anterior teeth are severely tilted lingually.



A: lingual bar and cingulum bar major connector: The upper portion of this major connector is located on the cingula of the anterior teeth. The requirement of positive support by rest seats, at least as far anteriorly as the canines, is critical. This type of major connector easily traps food and is often more objectionable to patients than a linguoplate.

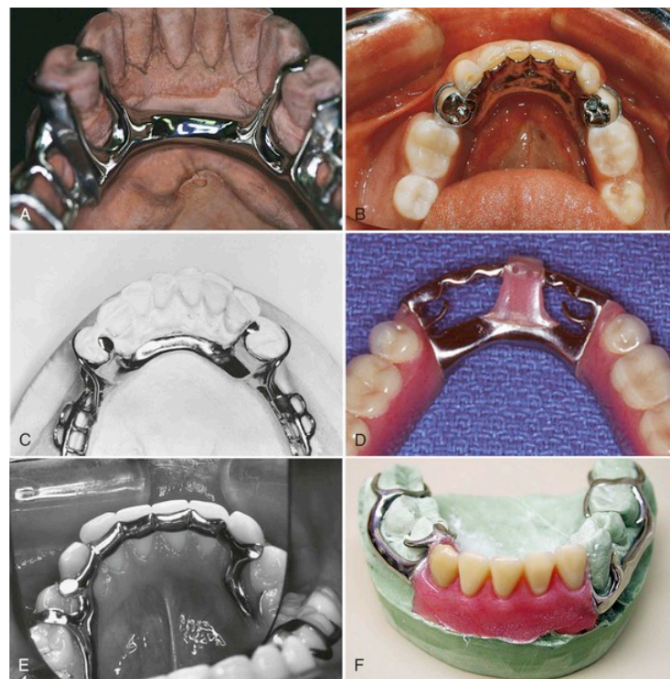
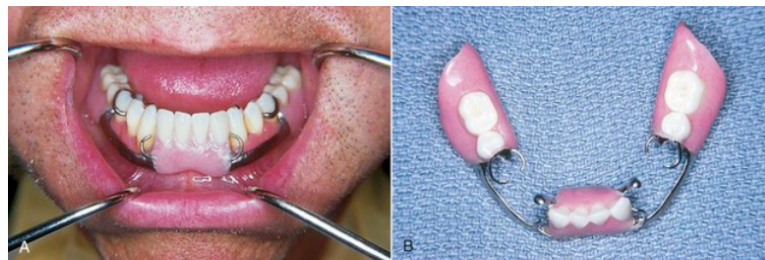
B: cingulum bar (continuous bar) major connector: Although this design may reduce the possibility of food entrapment, it may not provide adequate rigidity.

E. Labial bar

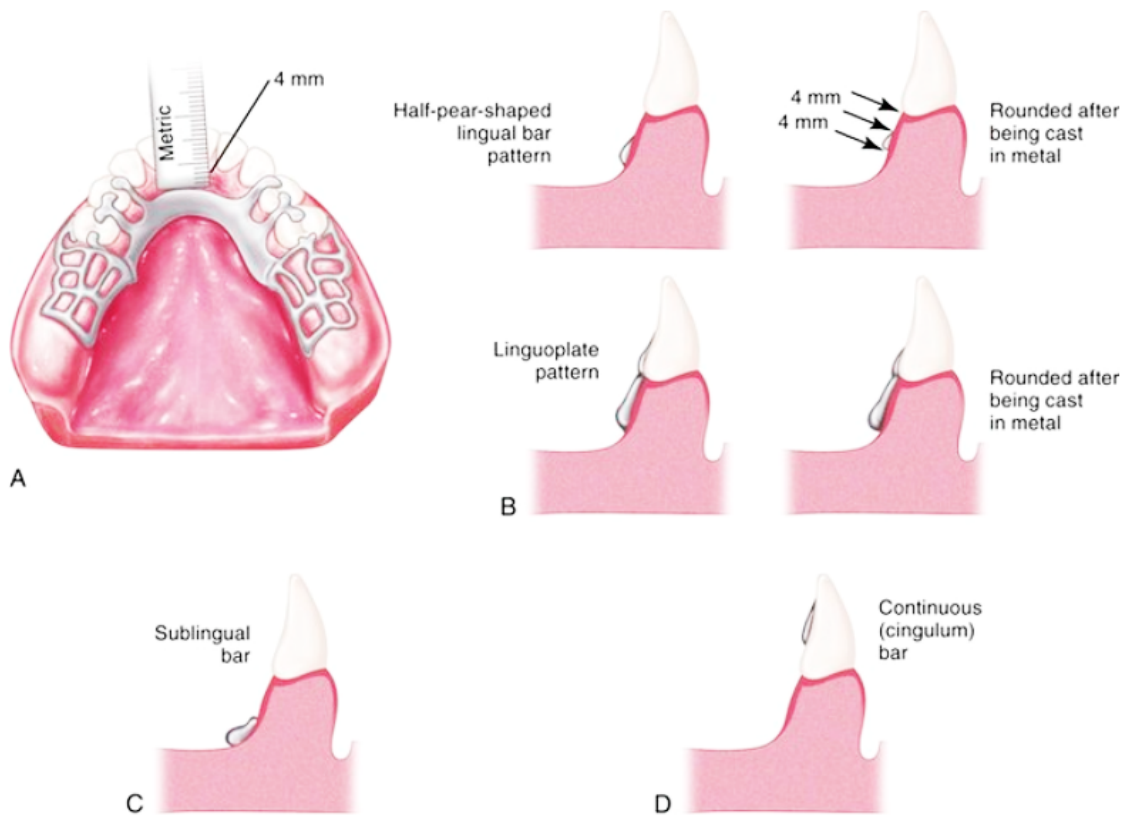
With the use of conservative mouth preparations in the form of recontouring and blockout, a lingual major connector can almost always be used. Lingually inclined teeth may sometimes have to be reshaped using crowns.

The same applies to the use of a labial bar when a mandibular torus interferes with the placement of a lingual bar. Unless surgery is contraindicated, interfering mandibular tori should be removed so that the use of a labial bar connector may be avoided.

Indications: In extreme lingual inclination of the remaining lower anterior teeth and premolars prevent the use of a lingual bar major connector.



Mandibular major connectors. A, Lingual bar. B, Linguoplate. C, Sublingual bar. D, Lingual bar with continuous bar (cingulum bar). E, Cingulum bar. F, Labial bar.



A, Lingual bar major connector should be located at least 4 mm inferior to the gingival margins and farther if possible. The vertical height of a finished lingual bar should be at least 4 mm for strength and rigidity. If less than 8 mm exists between the gingival margins and the movable floor of the mouth, a linguoplate (B), a sublingual bar (C), or a continuous bar (D) is preferred as a major connector. Relief is provided for soft tissue under all portions of the mandibular major connector and at any location where the framework crosses the marginal gingiva. The inferior border of the mandibular major connectors should be gently rounded after being cast to eliminate a sharp edge.

Methods for measuring the height of the floor of the mouth

Two clinically acceptable methods may be used to determine the relative height of the floor of the mouth and locate the inferior border of a lingual mandibular major connector.

1. The first method is to measure the height of the floor of the mouth in relation to the lingual gingival margins of adjacent teeth with a periodontal probe. When these measurements are taken, the tip of the patient's tongue should just lightly touch the vermilion border of the upper lip. Recording of these measurements permits their transfer to both diagnostic and master casts.
2. The second method is to use a special tray for which lingual borders are 3 mm short of the elevated floor of the mouth, and then to use an impression material that will permit the impression to be accurately moulded as the patient licks the lips. The inferior border of the planned major connector can then be located at the height of the lingual sulcus of the cast resulting from such an impression.

